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Definition: Random Definitions $\mathbb{N} = \mathbb{N}_0$, ie $\mathbb{N} = \{n \in \mathbb{Z} : n \geq 0\}$ - In this course we only use binary codes, ie $F = \mathbb{F}_2$, the two-element field - For $a, b \in \mathbb{Z}$, a divides b, or $a b$, if $\exists c \in \mathbb{Z}$ s.t. $b = ac$ $a, b \in \mathbb{Z}$ are relatively prime or coprime if $\gcd(a, b) = 1$
Theorem: Random Theorems Well-Ordering: All sets of positive ints $\neq \emptyset$ contain a least element
Definition: Primitive Pythagorean Triples A Pythagorean Triple is a set of numbers $(a, b, c) \in \mathbb{N}$ such that $a^2 + b^2 = c^2$. Note if (a, b, c) is a PT, so is (da, db, dc). A Primitive Pythagorean Triple is a PT such that a, b, c have no common factor. Such a, b, c have one of a, b odd and the other even. <i>Square of an odd is odd, square of an even is even. If both even then a, b, c divisible by 2. If both odd work through the eqn with $a = 2k+1, b = 2n+1$ to see all divisible by 4. Indeed, every PPT with (a, b, c) with a</i> odd has, for $s \geq t \geq 1$ odd coprime integers, $a = st, b = \frac{s^2 - t^2}{2}, c = \frac{s^2 + t^2}{2}$.
Theorem: The Division Algorithm Let $a, b \in \mathbb{Z}$ with $b > 0$. Then $\exists$ unique $q, r \in \mathbb{Z}$ s.t. $a = qb + r$ and $0 \leq r < b$. q is the quotient and r is the remainder.
Lemma: Bézout's Lemma Let a, b be nonzero integers with $\gcd(a, b) = d$. Then $\exists s, t \in \mathbb{Z}$ s.t. $as + bt = d$. Moreover any common divisor of a and b divides d.
Theorem: Fermat's Last Theorem For $n, x, y, z \in \mathbb{Z}$, and $n > 2$, there are no integer solutions to $x^n + y^n = z^n$.
Theorem: The Extended Euclidean Algorithm Let $a, b \in \mathbb{Z}$ be s.t. $a \geq b > 0$. Then $\gcd(a, b)$ can be found as follows: - By Division Algorithm $\exists$ unique q_1, r_1 s.t. $a = q_1 b + r_1$ with $0 \leq r_1 < b$ - If $r_1 > 0$, apply the Division Algorithm to b, r_1 for $b = q_2 r_1 + r_2$. - Repeat <i>ad infinitum</i>, the last non-zero remainder is $\gcd(a, b)$
Example: Finding GCD By Euclidean Algorithm Find $\gcd(112, 27)$. $112 = 4(27) + 4$ $27 = 6(4) + 3$ $4 = 1(3) + 1$ $3 = 3(1) + 0$ So $\gcd(112, 27) = 1$ and $112, 27$ are coprime
Definition: Congruence a is congruent to b mod n, written $a \equiv b \pmod n$, if $n (a - b)$, ie $\exists q \in \mathbb{Z}$ s.t. $a = nq + b$. For $a, b, c, d, n \in \mathbb{Z}$ with $n > 0$, $a \equiv b \pmod n$, $c \equiv d \pmod n$: $a + c \equiv b + d \pmod n$ $a - c \equiv b - d \pmod n$ $a \cdot c \equiv b \cdot d \pmod n$
Definition: System Of Residues A complete system of residues modulo n is a set $I \subseteq \mathbb{Z}$ s.t. every $k \in \mathbb{Z}$ is congruent mod n to exactly one $i \in I$
Theorem: Wilson's Theorem If p is prime then $(p-1)! \equiv -1 \pmod p$
Major Theorem: The Chinese Remainder Theorem Let $m_1, m_2, \dots, m_k \in \mathbb{N}$ be pairwise coprime. Then given $a_1, a_2, \dots, a_k \in \mathbb{Z}$, the system of congruences $\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$ has a unique soln mod N ($N = m_1 \cdot m_2 \cdot \dots \cdot m_k$). Constructive proof: <i>Proof.</i> Let $N_i = n/m_i$ be the product of all m_j with $j \neq i$. Since m_i is coprime to m_j for $i \neq j$, we must have $\gcd(N_i, m_i) = 1$. Then since N_i and m_i are coprime, N_i must have an inverse mod m_i, ie $\exists x_i \in \mathbb{Z}$ s.t. $N_i x_i \equiv 1 \pmod{m_i}$ (to find x_i use the Euclidean algorithm to find numbers s.t. $N_i x_i + m_i y_i = 1$). Then consider the product $a_i N_i x_i$, which is s.t. $a_i N_i x_i \equiv 0 \pmod{m_j}$ for $i \neq j$, and $a_i N_i x_i \equiv a_i \pmod{m_i}$. Thus $x = \sum_{i=1}^k a_i N_i x_i$ precisely satisfies $x \equiv a_i \pmod{m_i}$ for all i. $\square$
Example: Applying The CRT Solve the system of congruences $\begin{aligned} x &\equiv 1 \pmod 2 \\ x &\equiv 3 \pmod 5 \\ x &\equiv 4 \pmod 7 \end{aligned}$ We have $N = 2 \cdot 5 \cdot 7 = 70$, $N_1 = 5 \cdot 7 = 35$, $N_2 = 2 \cdot 7 = 14$, $N_3 = 2 \cdot 5 = 10$. Calculate by Euclidean Algorithm or inspection for: $\begin{aligned} 35 \cdot 1 &\equiv 1 \pmod 2 \\ 14 \cdot 4 &\equiv 1 \pmod 5 \\ 10 \cdot 5 &\equiv 1 \pmod 7 \end{aligned}$ so we have $x_1 = 1, x_2 = 4, x_3 = 5$. Thus $x \equiv (1 \cdot 35 \cdot 1) + (3 \cdot 14 \cdot 4) + (4 \cdot 10 \cdot 5) \equiv 403 \equiv 53 \pmod{70}$.

Theorem: Applying The CRT for Non-coprime $m_i m_j$ Key fact - if m, n are coprime integers, and $a \in \mathbb{Z}$, the congruence $x \equiv a \pmod{m \cdot n}$ has the same integer solutions as the system $\begin{aligned} x &\equiv a \pmod m \\ x &\equiv a \pmod n \end{aligned}$ Thus knowing $x \pmod N$ is equivalent to knowing $x \pmod{a_i}$ each of the primes in the prime decomposition of N, and to solve such systems we can simply decompose it to a larger system where all m_i are coprime.
Euler & Rings Major Definition: Euler's Function For $n > 1$, the Euler function $\phi(n)$ is defined to be the number of natural numbers not exceeding n which are coprime with n. $\phi(1) := 1$. Major Theorem: Euler's Theorem For $n \in \mathbb{N}$ s.t. $n > 1$, and $a \in \mathbb{Z}$ s.t. a, n coprime, then $a^{\phi(n)} - 1 \equiv 0 \pmod n$ Fermat's Little Thm: If p prime and x, p coprime; $x^{p-1} \equiv 1 \pmod p$ - If p prime and x, p coprime; $x^p \equiv x \pmod p$
Definition: Carmichael Numbers $n \in \mathbb{Z}$ is a Carmichael number if $a^{n-1} \equiv 1 \pmod n$ for all a coprime to n If n is s.t. $a^{n-1} = 1 \pmod n$ for all $0 < a < n$, then n is prime
Major Definition: Rings (Different Axioms To Halg!) A ring is a set R equipped with $+, \cdot : R \times R \rightarrow R$ s.t. for all $a, b, c \in R$: $(R, +)$ is an abelian group - Associativity of $\cdot$: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ - Distributivity: $a \cdot (b + c) = a \cdot b + a \cdot c$; and $(a + b) \cdot c = a \cdot c + b \cdot c$ - Variants of rings, with the above axioms true: - Commutative ring: $a \cdot b = b \cdot a$ - Unital ring: $\exists 1_R \in R$ s.t. $1_R \cdot a = a \cdot 1_R = a \forall a \in R$ - Integral domain: Ring with no zero-divisors ($x, y \in R$ s.t. $x \cdot y \neq 0, x \cdot y = 0$)
Definition: The Ring $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}_n$ The elements of $\mathbb{Z}_n$ are congruence classes mod n, ie $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$. We define the ring addition/multiplication by, for $0 \leq a, b < n$: $\overline{a} + \overline{b} = \overline{c}$ iff $a + b = c \pmod n$ $\overline{a} - \overline{b} = \overline{c}$ iff $a - b = c \pmod n$ $\overline{a} \cdot \overline{b} = \overline{c}$ iff $a \cdot b = c \pmod n$
Definition: Primitive Roots For prime $p > 0$, $i \in \mathbb{Z}$ is a primitive root modulo p iff $i, i^2, \dots, i^{p-1}$ are pairwise distinct mod p. Ie, $i^m \not\equiv 1 \pmod p$ for $0 < m < p-1$ Equivalent definitions: $i \in \mathbb{Z}$ is a primitive root modulo p iff: $p \nmid (i^j - 1)$ for any $0 < j < p-1$ and $p i^{p-1} - 1$. - For every $m \in \mathbb{Z}$ not divisible by p, $\exists a \in \mathbb{N}$ s.t. $m \equiv i^a \pmod p$.
Theorem: Primitive Root Theorem For every prime $p > 0$ there exists a primitive root modulo p. Let $x \in \mathbb{Z}$, $n \in \mathbb{N}$, and $\alpha, \beta \in \mathbb{N}$, $\beta > 0$. Suppose $x\alpha \equiv 1 \pmod n$ and $x\beta \equiv 1 \pmod n$. Then $x \gcd(\alpha, \beta) \equiv 1 \pmod n$
Major Definition: The Legendre Symbol Let $p > 0$ be prime and $a \in \mathbb{Z}$ s.t. a is not divisible by p. The Legendre symbol of a modulo p, denoted $(\frac{a}{p})$, is equal to 1 if $a = r^2 \pmod p$ for $r \in \mathbb{Z}$, and equal to -1 otherwise. If $(\frac{a}{p}) = 1$ then a is a quadratic residue modulo p.
Theorem: Legendre Symbol Theorems - Let p be an odd prime. Then $(\frac{a}{p}) \equiv a^{(p-1)/2} \pmod p$ $(\frac{a}{p})(\frac{b}{p}) = (\frac{ab}{p})$ $(\frac{a}{p}) = (\frac{a-p}{p})$ and moreover if $a \equiv b \pmod p$ then $(\frac{a}{p}) = (\frac{b}{p})$ - If $p, q > 0$ are distinct odd primes, and $p \equiv 1 \pmod 4$ or $q \equiv 1 \pmod 4$ then $(\frac{p}{q}) = (\frac{q}{p})$ - If $p, q > 0$ are distinct odd primes, and $p \equiv 3 \pmod 4$ and $q \equiv 3 \pmod 4$, then $(\frac{p}{q}) = -(\frac{q}{p})$
Lemma: The Gauss Lemma Let p be an odd prime and $a \in \mathbb{Z}$ not divisible by p. Consider $S := \{a, 2 \cdot a, 3 \cdot a, \dots, \frac{p-1}{2} \cdot a\}$. Let S' be the set of remainders of elements of S when divided by p, with each remainder in the interval $(-\frac{p}{2}, \frac{p}{2})$. Let n be the number of elements in S' smaller than 0. Then $(\frac{a}{p}) = (-1)^n$ For prime $p > 2$, $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$

Major Definition: Gaussian Integers The Gaussian integers are the elements of $\mathbb{Z}[i] := \{a + ib : a, b \in \mathbb{Z}\}$. $\mathbb{Z}[i]$ is closed under $+/ -$ thus is a ring. - For $w, z \in \mathbb{Z}[i]$, $z = a + bi = \sqrt{a^2 + b^2}$ and $z \cdot w = z \cdot w$ $z \in \mathbb{Z}[i]$ is a unit if $\exists z' = a' + ib' \in \mathbb{Z}[i]$ s.t. $z \cdot z' = 1$. For all units $z \in \mathbb{Z}[i]$, $z = 1$ and $z \in \{\pm 1, \pm i\}$ - For $z, z' \in \mathbb{Z}[i]$, $z \neq 0$, we say z divides z', written $z z'$, iff $z' = z \cdot w$ for some $w \in \mathbb{Z}[i]$ - Let $z \in \mathbb{Z}[i]$ be non-zero and not a unit. z is an irreducible Gaussian integer if whenever $z = u \cdot v$ for $u, v \in \mathbb{Z}[i]$, either u or v is a unit. - Let $z \in \mathbb{Z}[i]$ be non-zero and not a unit. z is a prime Gaussian integer if whenever $z (u \cdot v)$ for $u, v \in \mathbb{Z}[i]$, $z u$ or $z v$. $z \in \mathbb{Z}[i]$ is prime if and only if it is irreducible
Theorem: Division Algorithm For Gaussian Integers Let $z, w \in \mathbb{Z}[i]$ with $w \neq 0$. Then $\exists q, r \in \mathbb{Z}[i]$ with $z = qw + r$ and $r < w$.
Definition: GCD For Gaussian Integers Let $a, b \in \mathbb{Z}[i]$ not both zero. Then d, the greatest common divisor of a and b, is a Gaussian integer which divides both of them, and any other divisor of a and b divides d.
Lemma: Bézout's Lemma For Gaussian Integers Let $a, b \in \mathbb{Z}[i]$ not both zero. Then d, the greatest common divisor of a and b, exists, and so do $s, t \in \mathbb{Z}[i]$ s.t. $as + bt = d$.
Theorem: Writing Integers As A Sum Of Two Squares Let $p \equiv 1 \pmod 4$ be a prime. Then $p = a^2 + b^2$ for some $a, b \in \mathbb{Z}$. If prime $p \equiv -1 \equiv 3 \pmod 4$, then p is a prime Gaussian integer and $\nexists a, b \in \mathbb{Z}$ s.t. $p = a^2 + b^2$ Let $1 < n, n \in \mathbb{N}$. Suppose $n = 2^c p_1^{a_1} \dots p_k^{a_k} q_1^{d_1} \dots q_t^{d_t}$ is the prime factorisation of n into positive primes, with each $p_i \equiv 1 \pmod 4$ and each $q_j \equiv 3 \pmod 4$ where $c, k, t \geq 0, \in \mathbb{Z}$ and $a_1, \dots, a_k, d_1, \dots, d_t > 0 \in \mathbb{Z}$. Then $\exists a, b \in \mathbb{Z}$ with $n = a^2 + b^2$ iff each d_i is even.
Definition: Associate Integral Domain Elements Let R be an integral domain and $a, b \in R$. We say a is an associate of b if $a = bu$ for u a unit in R. If a is an associate of b, b is an associate of a.
Theorem: The FTA For Gaussian Integers Let $a \in \mathbb{Z}[i]$ be nonzero. Then either a is a unit or a may be expressed as a product of finitely many primes. Moreover, if $a = p_1 \dots p_k = q_1 \dots q_t$ where all p_i, q_j are prime Gaussian integers then the p_i and q_j can be paired such that paired primes are associates.

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Major Definition: Linear Binary Codes A binary word is a sequence of 1s and 0s. The word length is the number of digits in the word. A binary code is a set of binary words. A code C is linear iff for all $y, z \in C$, $x + y \in C$. Binary words which are elements of a binary code are codewords. We can ascribe meaning to codewords (eg $(1, 1, 1)$ = 'yes' and $(0, 0, 1)$ = 'no'). A code is a block code if all words have the same length. A repetition code of length n is obtained by repeating $n-1$ times the first single symbol. Example of length 3: $\{(1, 1, 1), (0, 0, 0)\}$. - A binary linear code has exactly 2^k elements. - Binary linear codes can be considered as linear spaces over $\mathbb{F} = \{0, 1\}$. $x \cdot y := \sum_{i=1}^n x_i y_i$. $(\cdot)$ is a symmetric bilinear form on $\mathbb{F}_2^n$. x and y are orthogonal if $x \cdot y = 0$. - A parity check bit is a bit appended to a codeword to make the sum of the digits even. All binary codewords with such bits are of form $(x_1, \dots, x_{n-1}, \sum_{i=1}^{n-1} x_i)$
Definition: The Parameters Of A Linear Code A triple (n, k, d) gives the three params of a linear code - n, the length of codewords in the code; k, the dimension of the code $\dim_{\mathbb{F} C}$, and d the minimal distance of the code. k is equal to the number of linearly independent codewords in the code - There are no linear binary codes with params (n, k, d) where $n < k + d - 1$
Definition: Neighbourhood Of A Word For a binary word $x = (x_1, \dots, x_n)$, with all $x_i \in \{0, 1\}$ and $r \geq 0$, the neighbourhood of x with radius r is the set $N_r(x) = \{y = (y_1, \dots, y_n) : y_1, \dots, y_n \in \{0, 1\}, d(x, y) \leq r\}$. Note that $x \in N_r(x)$ and $N_r(x) = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}$
Definition: Hamming Distance For C a code over $\mathbb{F}_2$, $x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n) \in C$; the Hamming distance d on C is $d(x, y) := \{i \in \{1, \dots, n\} : x_i \neq y_i\}$ - For $c \in C$, the weight of c is $\text{wt}(c) := d(c, 0) = \{i \in \{1, \dots, n\} : c_i \neq 0\}$ - For C, f binary codes of the same length, $\text{wt}(c + f) = \text{wt}(c) + \text{wt}(f) - 2\text{wt}(cf)$ - The minimal distance of C is $d(C) = \min\{d(x, y) : x, y \in C, x \neq y\}$, intuitively the smallest distance between any two codewords in the code. - For C a linear code, the minimal distance is equal to the minimum weight among non-zero codewords in the code. - For binary words x, y, z of length n; $d(x, y) + d(y, z) \geq d(x, z)$

Definition: Decision Rules For $C \subseteq F^n$ a code, a decision rule for C is a function $\sigma : F^n \rightarrow C$ which assigns to each $z \in F^n$ a codeword $c \in C$.
Definition: Bit-Error A bit-error is a mistake in transmission of a codeword z in a code C, where one digit of z is flipped (either from 0 to 1, or from 1 to 0). A code is r-error-correcting if it has minimal distance $d \geq 2r + 1$.
Theorem: Minimal Distance Rule For C a code, the minimal distance rule states that given a codeword $z \in C$ received through a noisy channel, the receiver should choose $\sigma(z)$ to be the codeword $c \in C$ such that $d(z, c)$ is minimised. If a sender uses a code C with minimum distance $d > 2r$ and the receiver uses the MD rule, provided no more than r bit-errors are made in transmitting any codeword, the MD rule will map every received codeword to the correct transmitted one.
Definition: A Generator Matrix Of A Code Let C be a linear code with params (n, k, d). Since C has $\dim_{\mathbb{F}} C = k$ we can find codewords $c_1, \dots, c_k$ which form a basis of C. Then the $k \times n$ matrix with rows $c_1, \dots, c_k$ is a generator matrix of C
Definition: The Dual Code For C a code with params (n, k, d), the dual code $C^\perp = \{x = (x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbb{F}_2, x \cdot c = 0 \forall c \in C\}$ is the orthogonal complement of C. $\dim_{\mathbb{F}} C^\perp = n - k$ since $\dim_{\mathbb{F}} C = k$. - A code is self-dual if $C = C^\perp$, for example $C = \{(0, 0), (1, 1)\}$. $(C^\perp)^\perp = C$. $\dim C + \dim C^\perp = n$.
Definition: Parity Check Matrices For C a code with params (n, k, d), a generator matrix H for $C^\perp$ (a matrix whose rows form a basis of $C^\perp$) is a parity check matrix for C. H is a $(n - k) \times n$ matrix c is a codeword in C iff $Hc^T = 0$ iff $cH^T = 0$. C has minimal distance d if and only if H has d lin-dep cols and does not have $d-1$ lin-dep cols
Definition: Perfect Codes For C a binary code of length n, minimal distance $d = 2r + 1$ for $r \geq 0 \in \mathbb{Z}$. C is a perfect code iff $C \left(\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r} \right) = 2^n$, or equivalently if each $c \in C$ of length n is in exactly one neighbourhood $N_r(c)$
Theorem: Adjusting Code Parameters Let C be a linear binary code with parameters (n, k, d) - If $k > 1$; $\exists$ a lin-bin code with params $(n-1, k', d')$; $k' \geq k-1, d' \geq d$. - If d is odd, $\exists$ a lin-bin code with params (n, k', d'); $k' \geq k-1, d' > d$. - If $d > 1$, there is a lin-bin code with params $(n, k, d-1)$
Theorem: More Code Param Theorems - Let $r, d \in \mathbb{Z}$ s.t. $r \geq 0, d \geq 2r + 1$, and $n \in \mathbb{N}$. Then there are no codes C of length n s.t. $2^n < 2^k \left(\binom{n}{0} + \dots + \binom{n}{r} \right)$ - Gilbert-Varshamov: Let $r, n \in \mathbb{N}, r \geq 0$. Then $\exists$ a binary code C of length n with minimal distance $d \geq 2r + 1$ s.t. $C \left(\binom{n}{0} + \dots + \binom{n}{2r} \right) \geq 2^n$.